

Comprehensive Project Deep-Dive & Onboarding Manual

Executive Overview, Architectural Breakdown, Data Pipelines, and Research Roadmap

Target Audience: New Core Engineers & Researchers

Project Version: v1.0 Production Readiness

Document Date: August 2026

Scope: Mobile App, Web Portal, ML Engine & Backend

1. Non-Technical Executive Overview

What is AlloteMe?

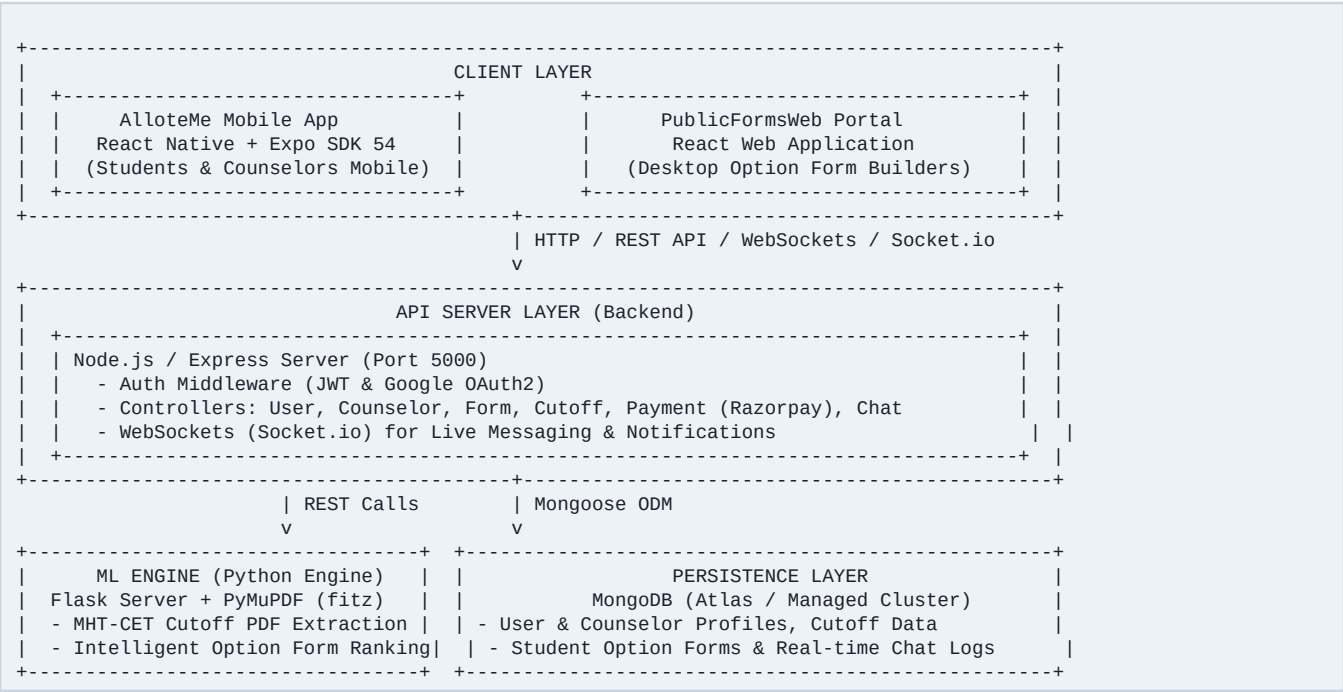
AlloteMe (integrated with *CounselMe*) is a centralized counseling and college preference allotment platform. Every year, thousands of students undergo engineering entrance examinations (such as MHT-CET, JEE Main). The allotment process requires submitting complex choice lists (Option Forms), where students prioritize college branches based on category cutoffs, location, fees, and historical seat matrix distributions. A single error in preference ordering can lead to missing seat allocation entirely.

Core Value Proposition & Real-World Solutions:

- **Smart Option Form Generator:** Automatically generates tailored college priority lists based on student rank, percentile, caste category, and branch preferences, eliminating manual human errors.
- **Expert Counselor Ecosystem:** Connects students directly with expert human counselors who review, edit, and lock option forms via dedicated mobile and web tools.
- **Automated PDF Document Parsing:** Extracts multi-stage cut-off matrices directly from official government PDF booklets using Machine Learning routines.
- **Real-Time Live Chat & Notifications:** Ensures seamless, instant communication between counselors, parents, and students with real-time updates and push alerts.
- **Unified Mobile & Web Ecosystem:** Cross-platform accessibility allowing students to track their counseling lifecycle on iOS, Android, or desktop web browsers.

2. Platform High-Level Architecture

The AlloteMe ecosystem operates on a micro-service-oriented architecture comprising four primary tiers: Client Layer, API Server Layer, Intelligent ML Engine, and Persistence Layer.



3. Technical Deep Dive by Subsystem

3.1 Mobile & Web Frontend Tier

Technology Stack: React Native, Expo SDK 54, React Navigation v7, React Native Web.

Key Functionalities: Cross-platform build target covering Android (APK via EAS), iOS, and Web. Complex interactive UI supporting drag-and-drop preference reordering (using *@dnd-kit* and *react-native-draggable-flatlist*). Native feature integrations: Expo Camera, Document Picker, PDF Generator (Expo Print), Secure Store, and Razorpay SDK for payments.

3.2 Node.js Backend Microservices

Technology Stack: Node.js, Express.js, MongoDB (Mongoose ODM), Socket.io, JWT Authentication.

Core Models & Responsibilities:

Model / Module	Database Entity Responsibilities	Key Features
User.js	Stores student profiles, entrance ranks, category, assigned counselor, and auth tokens.	Google OAuth, JWT Auth, Role management.
Counselor.js	Profile of verified counselors, assigned students, pricing tiers, and rating metrics.	Counselor verification, capacity & review tracking.
Form.js	Student option form entries, ordered college choices, status (Draft/Locked/Submitted).	Version control, counselor approval locking flow.
Cutoff.js	College information, DTE codes, branch codes, category-wise percentile ranks.	Indexed fast search across college branches.
Chat.js	Real-time messaging history between students and assigned counselors.	Socket.io room handling, instant message sync.

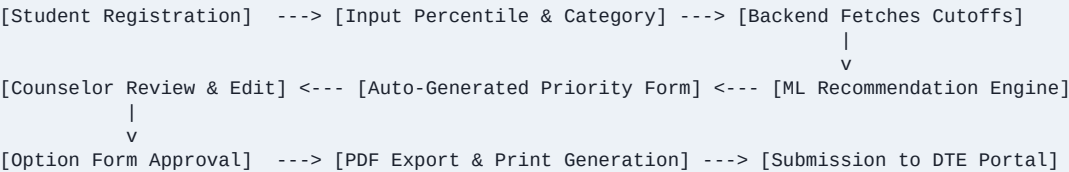
3.3 Machine Learning & Data Extraction Engine

Technology Stack: Python 3, Flask, PyMuPDF (fitz), Regular Expressions (re), Flask-CORS.

Parsing Unstructured Cut-off PDFs:

- 1. **Header Extraction:** Scans text blocks for 4-5 digit DTE codes (e.g., 6007 - Walchand College of Engineering).
- 2. **Branch Identification:** Detects 9-12 alphanumeric course codes (e.g., 600724510 - Computer Science).
- 3. **Matrix Alignment:** Uses coordinate spatial sorting (b[1], b[0]) to group category names (GOPENH, LOPENH, TFWS, PWD) with rank and percentile tuples.
- 4. **JSON Export:** Converts complex PDF tables into structured JSON payloads for MongoDB storage.

4. End-to-End User Lifecycle & Data Flow



5. Research & Optimization Roadmap for New Team Members

To make the AlloteMe project significantly more efficient and useful, incoming team members should prioritize research in these domain areas:

Research Focus Area	Current State	Proposed Research Goal & Innovation
1. Advanced Predictive ML Models	Rule-based cutoff matching against prior year data.	Implement ML models (XGBoost/Random Forest) to predict <i>probabilistic seat allocation</i> taking seat matrix shifts and historical percentile trends into account.
2. AI Counselor Chatbot	Direct human counselor chat via Socket.io.	Integrate an LLM assistant (RAG over DTE brochures) to answer routine student queries 24/7, reducing human counselor workload.
3. Document Parser Optimization	Python PyMuPDF string regex parser in Flask.	Research Computer Vision table extraction (PaddleOCR, Camelot) to handle multi-column scanned PDF cut-off sheets without structural layout failures.
4. Performance & Caching	Direct MongoDB queries per filter request.	Introduce Redis caching for read-heavy cut-off data queries, reducing database load by over 80% during peak CAP round result releases.
5. Multi-Tenant Counseling Portal	Single-institution / centralized database.	Scale the platform into a SaaS framework where third-party coaching institutes can brand and manage their own student counseling cohorts.

Document Summary: Prepared for internal team onboarding. For questions or setup assistance, consult the repository codebase under CounselMe/backend, ML/app.py, and root React Native mobile workspace.